CSS 324 - Introduction to Machine Learning

Machine Learning for Animal Disease Detection

A Multispecies Diagnostic AI System

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Abstract

Traditional animal disease diagnosis methods often depend on manual examination by veterinarians, which can be time-consuming and difficult when dealing with many different animal species. In this project, a machine learning system was developed to predict animal diseases using clinical symptoms, vital signs, and health indicators from multiple species.

In this project several preprocessing and feature engineering techniques were applied, missing values, text-based measurements, and categorical variables were handled, and additional features were created to improve model performance. Three machine learning models were trained and compared: Logistic Regression, Random Forest, and XGBoost.

In general, this project demonstrates how machine learning can support veterinary diagnostics and improve early disease detection among multiple animal species.

1 Introduction

Animal health plays an important role in agriculture, wildlife conservation, zoos, and veterinary medicine. Early detection of diseases can reduce animal mortality, prevent disease outbreaks, and improve treatment outcomes. However, diagnosing diseases in animals can be difficult because symptoms often vary across species, and many diseases share similar clinical signs.

Machine learning allows us to improve disease detection by analyzing large amounts of medical and behavioral data automatically. By learning patterns from previous cases, ML models can help veterinarians in identifying possible diseases faster and more consistently.

The goal of this project was to build a multispecies diagnostic machine learning system capable of predicting animal diseases from clinical symptoms and vital signs. The project also aimed to explore how feature engineering and species-specific health indicators could improve prediction accuracy.

Main objectives:

- Analyze multispecies clinical records;
- Perform data cleaning and preprocessing;
- Create health-related features;
- Train and compare multiple ML models;
- Develop an interactive prediction system for disease diagnosis.

This project demonstrates the potential of ML in veterinary healthcare and highlights the strengths and limitations of disease prediction systems based on clinical symptoms alone.

2 Literature Review

Machine learning is increasingly used in healthcare and veterinary medicine to support disease diagnosis and health monitoring. Classification algorithms such as Logistic Regression, Random Forest, and

Gradient Boosting are commonly applied to medical datasets because they can identify patterns in clinical data and assist with early diagnosis.

Recent studies have also shown that artificial intelligence can improve animal health monitoring in agriculture and veterinary systems. Machine learning models can help detect abnormal health conditions, monitor animal behavior, and support veterinarians in making faster decisions. However, disease prediction remains difficult because many diseases share similar symptoms.

This project applies these ideas to a multispecies veterinary dataset by using machine learning models and species-specific health indicators to predict animal diseases.

3 Dataset Description

The project used the **Multispecies Diagnostic Records** dataset, which contained veterinary information for 14,976 animals across 32 species. The dataset included animal characteristics, vital signs, symptoms, and disease diagnoses.

The target variable was `Disease_Prediction`, which contained seven classes:

- Healthy
- Heat Stress
- Parasitic Gastritis
- Viral Infection
- Bacterial Pneumonia
- Nutritional Deficiency
- Fungal Dermatitis

The "Healthy" category had the largest number of records, while the disease classes were relatively balanced. Some columns contained missing values and text-based measurements, which required preprocessing before training the models.

4 Data Cleaning and Preprocessing

Several preprocessing steps were performed to prepare the dataset for machine learning.

Missing symptom values were treated as the absence of symptoms because healthy animals naturally had no reported symptoms. Numerical missing values were filled using median values, while missing categorical values were replaced with "Unknown."

Text-based measurements such as:

- "234.6 kg"

- "39.5°C"
- "2 weeks"

were converted into numerical values. Duration values were transformed into days to create a consistent format.

Categorical variables were encoded using:

- binary encoding for Yes/No symptoms
- Label Encoding for disease classes
- one-hot encoding for animal species

The dataset was then scaled using `StandardScaler` and divided into:

- 80% training data
- 20% testing data

The final machine learning dataset contained 53 features.

5 Feature Engineering

Several additional features were created to improve model performance and better represent animal health conditions.

The feature `Num_Symptoms` counted the total number of symptoms reported for each animal, while `Num_Clinical_Signs` summarized the number of abnormal clinical observations.

A custom `Health_Score` feature was also created:

$$Health\_Score = 10 - Num\_Symptoms - 0.5 \times Num\_Clinical\_Signs$$

Higher values indicated healthier animals.

Another important feature was species-specific fever detection. Since different animals have different normal temperatures, the project calculated each species' average temperature and measured how much an animal's temperature exceeded that average.

Animals with temperatures more than 2°C above their species average were classified as fever cases.

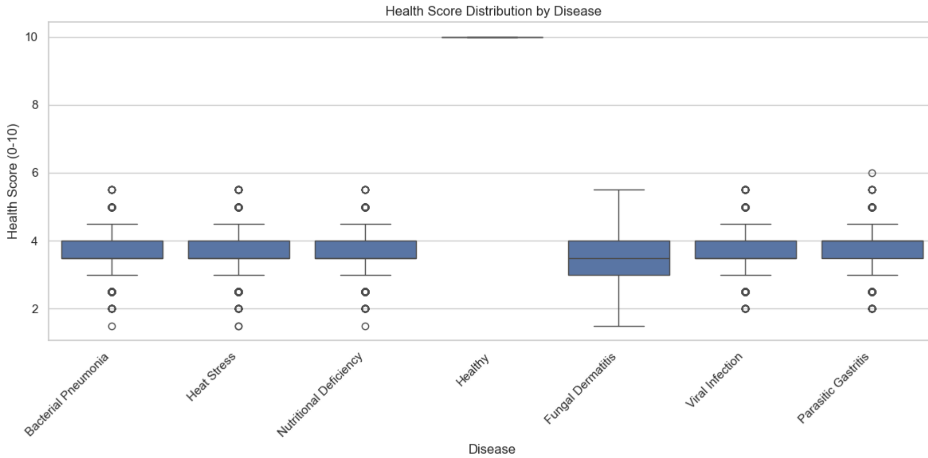

Figure 1: Feature Engineering Process

6 Exploratory Data Analysis (EDA)

EDA was performed to identify patterns in the dataset before training the models.

The analysis showed that healthy animals were the largest group in the dataset, while the disease classes had relatively similar distributions.

One important finding was that sick animals had higher body temperatures than healthy animals. On average, diseased animals were approximately 2.4°C hotter.

Symptom heatmaps also showed that many diseases shared similar symptoms such as coughing, vomiting, and appetite loss. This explained why predicting the exact disease was more difficult than identifying whether an animal was healthy or sick.

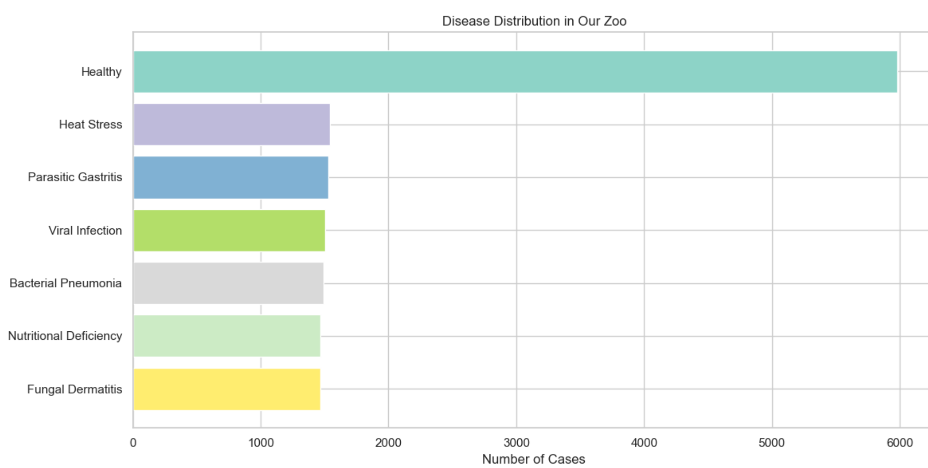

Figure 2: Disease Distribution

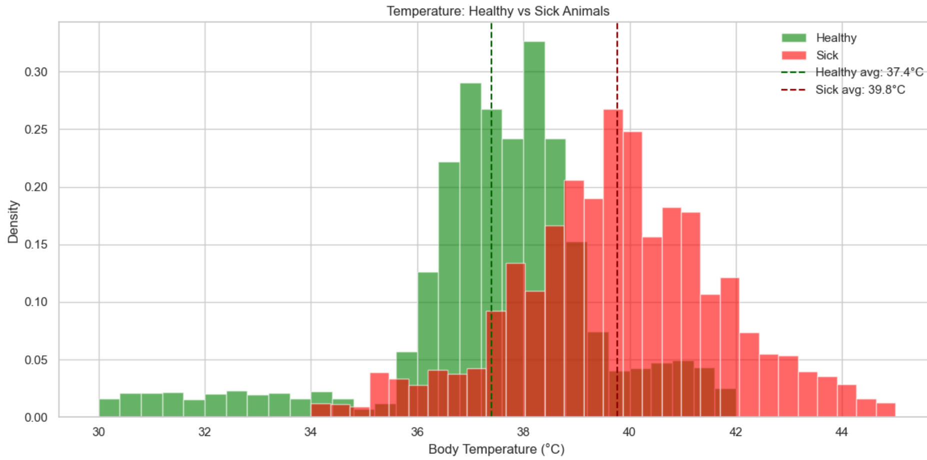

Figure 3: Temperature Comparison Between Healthy and Sick Animals

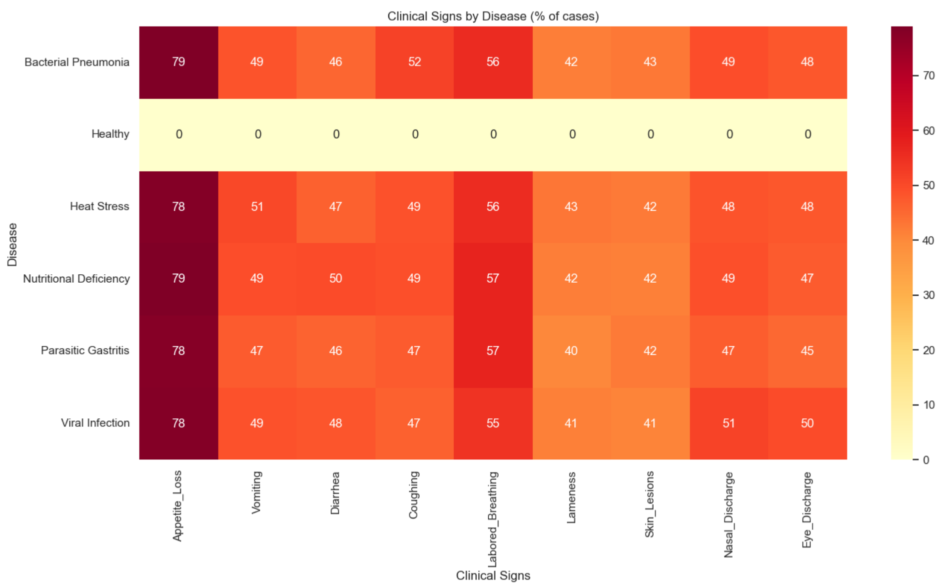

Figure 4: Clinical Signs by Disease

7 Machine Learning Models

Three machine learning models were trained and compared:

- Logistic Regression
- Random Forest
- XGBoost

The dataset was standardized using `StandardScaler` before training. Logistic Regression was used as a baseline model, while Random Forest and XGBoost were selected because of their strong performance on tabular datasets.

Table 1: Model Accuracy Comparison

Model	Accuracy
Logistic Regression	51.20%
Random Forest	49.87%
XGBoost	49.77%

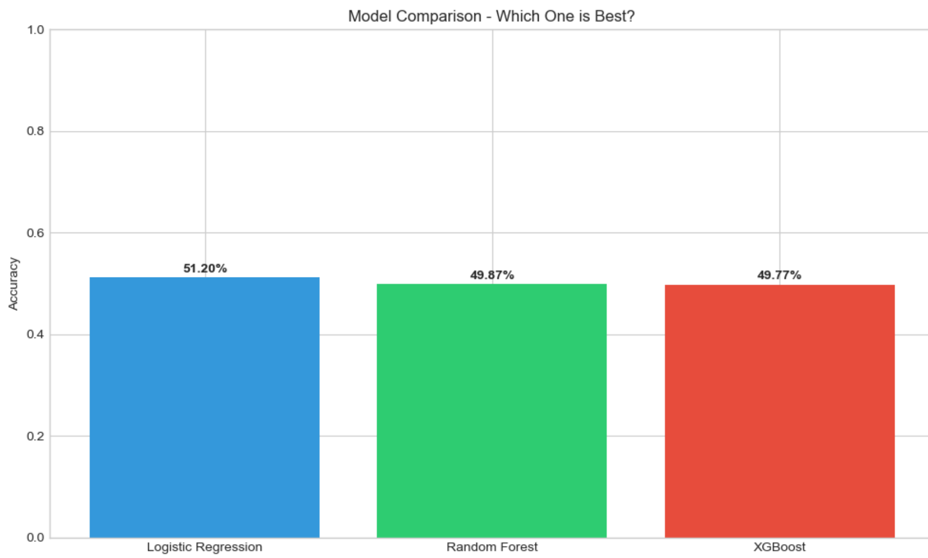

Figure 5: Model Accuracy Comparison

8 Results and Evaluation

The Logistic Regression model achieved an overall accuracy of 51.20%.

The model performed very well when identifying healthy animals, achieving near-perfect precision and recall for the "Healthy" class. However, the performance for specific diseases was lower because many diseases shared similar symptom patterns.

The confusion matrix showed that disease categories were frequently misclassified as one another, especially diseases with overlapping symptoms such as coughing, vomiting, and appetite loss.

Despite moderate accuracy, the project successfully demonstrated that machine learning can support veterinary diagnosis and distinguish healthy animals from sick animals using clinical data alone.

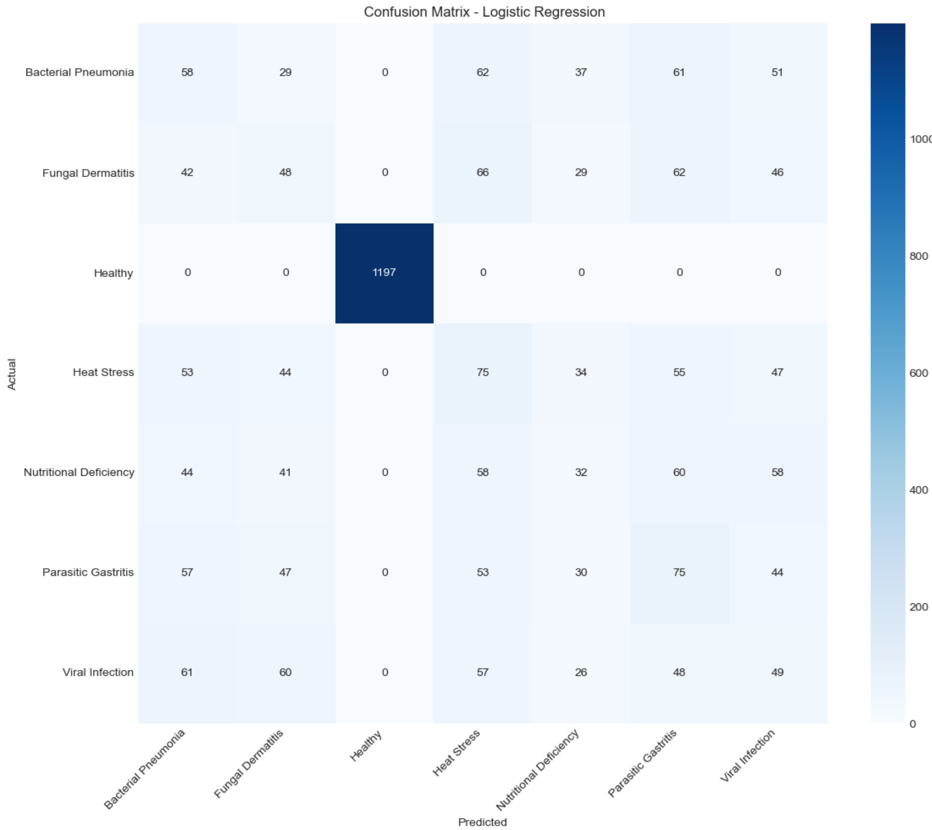

Figure 6: Confusion Matrix for Logistic Regression

9 Interactive Disease Prediction System

An interactive prediction function was developed to allow users to enter animal information and receive disease predictions in real time. The system used the same preprocessing and feature engineering pipeline as the training stage.

Two demonstration cases were tested:

- a sick lion with fever and respiratory symptoms
- a healthy penguin with normal vital signs

The project also included a simple veterinary diagnostic simulation game, showing how machine learning systems can support education and interactive learning.

10 Conclusion and Future Work

This project developed a multispecies machine learning system for predicting animal diseases using symptoms, vital signs, and engineered health indicators.

The project combined preprocessing, feature engineering, exploratory analysis, machine learning, and an interactive prediction system into one complete workflow. Logistic Regression achieved the best performance with 51.20% accuracy.

One of the main findings was that detecting whether an animal is healthy or sick is easier than identifying the exact disease because many diseases share similar symptoms.

In the future, the system could be improved by adding:

- larger datasets
- laboratory and imaging data
- deep learning models
- live sensor monitoring
- mobile or web-based veterinary applications

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